

UCS553: Enterprise Web Application

Lab Experiment 1

Basic programming using JAVA

1. Write Java Program to Check Leap Year

```
1 // Write Java Program to Check Leap Year
2
3 import java.util.Scanner;
4
5 public class LeapYear{
6     Run | Debug
7     public static void main(String[] args){
8         Scanner sc = new Scanner(System.in);
9
10        System.out.print(s: "Enter a year: ");
11        int year = sc.nextInt();
12
13        if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0)){
14            System.out.println(year + " is a Leap Year.");
15        }
16        else{
17            System.out.println(year + " is not a Leap Year.");
18        }
19        sc.close();
20    }
```

```
Enter a year: 2028
2028 is a Leap Year.
```

2. Write Java Program to Check Whether a Character is Alphabet or Not

```
1 // Write Java Program to Check Whether a Character is Alphabet or Not
2
3 import java.util.Scanner;
4
5 public class CheckAlphabet{
6     Run | Debug
7     public static void main(String[] args){
8         Scanner sc = new Scanner(System.in);
9
10        System.out.print(s: "Enter a character: ");
11        char c = sc.next().charAt(index: 0);
12
13        if ((c >= 'A' && c <= 'Z') || (c >= 'a' && c <= 'z')){
14            System.out.println(x: "It is an alphabet.");
15        }
16        else{
17            System.out.println(x: "It is not an alphabet.");
18        }
19        sc.close();
20    }
```

```
Enter a character: a
It is an alphabet.
```

3. Write Java Program to Find Factorial of a Number

```
1 // Write Java Program to Find Factorial of a Number
2
3 import java.util.Scanner;
4
5 public class Factorial{
6     public static void main(String[] args){
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print(s: "Enter a number: ");
10        int n = sc.nextInt();
11
12        int factorial = 1;
13
14        for (int i = 1; i <= n; i++){
15            factorial = factorial * i;
16        }
17
18        System.out.println("Factorial of " + n + " = " + factorial);
19
20        sc.close();
21    }
22 }
```

```
Enter a number: 5
Factorial of 5 = 120
```

4. Write Java Program to Display Fibonacci Series

```
1 // Write Java Program to Display Fibonacci Series
2
3 import java.util.Scanner;
4
5 public class Fibonacci {
6     public static void main(String[] args){
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print(s: "Enter the number of terms: ");
10        int n = sc.nextInt();
11
12        int first = 0, second = 1;
13
14        System.out.println(x: "Fibonacci Series:");
15
16        for (int i = 1; i <= n; i++) {
17            System.out.print(first + " ");
18
19            int next = first + second;
20            first = second;
21            second = next;
22        }
23        sc.close();
24    }
25 }
```

```
Enter the number of terms: 5
Fibonacci Series:
0 1 1 2 3
```

5. Write Java Program to Find GCD of two Numbers

```
1 // Write Java Program to Find GCD of two Numbers
2
3 import java.util.Scanner;
4
5 public class GCD{
6     public static void main(String[] args){
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print(s: "Enter first number: ");
10        int a = sc.nextInt();
11
12        System.out.print(s: "Enter second number: ");
13        int b = sc.nextInt();
14
15        while(b != 0){
16            int temp = b;
17            b = a % b;
18            a = temp;
19        }
20
21        System.out.println("GCD = " + a);
22
23        sc.close();
24    }
25 }
```

```
Enter first number: 4
Enter second number: 16
GCD = 4
```

6. Write Java Program to Find LCM of two Numbers

```
1 // Write Java Program to Find LCM of two Numbers
2
3 import java.util.Scanner;
4
5 public class LCM{
6     public static void main(String[] args){
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print(s: "Enter first number: ");
10        int a = sc.nextInt();
11
12        System.out.print(s: "Enter second number: ");
13        int b = sc.nextInt();
14
15        int lcm = (a * b) / findGCD(a, b);
16        System.out.println("LCM = " + lcm);
17        sc.close();
18    }
19
20    static int findGCD(int a, int b) {
21        while (b != 0) {
22            int temp = b;
23            b = a % b;
24            a = temp;
25        }
26        return a;
27    }
28 }
```

```
Enter first number: 4
Enter second number: 16
GCD = 4
```

7. Write Java Program to Count Number of Digits in an Integer

```
1 // Write Java Program to Count Number of Digits in an Integer
2
3 import java.util.Scanner;
4
5 public class CountDigits {
6     Run | Debug
7     public static void main(String[] args) {
8         Scanner sc = new Scanner(System.in);
9
10        System.out.print(s: "Enter an integer: ");
11        int n = sc.nextInt();
12
13        int count = 0;
14
15        while (n != 0) {
16            n = n / 10;
17            count++;
18        }
19
20        System.out.println("Number of digits = "+count);
21        sc.close();
22    }
23 }
```

```
Enter an integer: 10
Number of digits = 2
```

8. Write Java Program to Reverse a Number

```
1 // Write Java Program to Reverse a Number
2
3 import java.util.Scanner;
4
5 public class ReverseNumber{
6     Run | Debug
7     public static void main(String[] args){
8         Scanner sc = new Scanner(System.in);
9
10        System.out.print(s: "Enter a number: ");
11        int n = sc.nextInt();
12        int reverse = 0;
13
14        while (n != 0) {
15            int digit = n % 10;
16            reverse = reverse * 10 + digit;
17            n = n / 10;
18        }
19        System.out.println("Reversed number = " + reverse);
20        sc.close();
21    }
22 }
```

```
Enter a number: 900
Reversed number = 9
```

9. Write Java Program to Calculate the Power of a Number

```
1 // Write Java Program to Calculate the Power of a Number
2
3 import java.util.Scanner;
4
5 public class Power{
6     public static void main(String[] args){
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print(s: "Enter the base: ");
10        int base = sc.nextInt();
11        System.out.print(s: "Enter the exponent: ");
12        int exponent = sc.nextInt();
13
14        int result = 1;
15
16        for (int i = 1; i <= exponent; i++){
17            result = result * base;
18        }
19
20        System.out.println(base + " raised to the power " + exponent + " = " + result);
21        sc.close();
22    }
23 }
```

```
Enter the base: 2
Enter the exponent: 2
2 raised to the power 2 = 4
```

10. Write Java Program to Check Palindrome

```
1 // Write Java Program to Check Palindrome
2
3 import java.util.Scanner;
4
5 public class Palindrome {
6     public static void main(String[] args){
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print(s: "Enter a number: ");
10        int n = sc.nextInt();
11        int original = n;
12        int reverse = 0;
13
14        while(n != 0){
15            int digit = n % 10;
16            reverse = reverse * 10 + digit;
17            n = n / 10;
18        }
19
20        if(original == reverse){
21            System.out.println(x: "It is a Palindrome.");
22        }
23        else{
24            System.out.println(x: "It is not a Palindrome.");
25        }
26        sc.close();
27    }
28 }
```

```
Enter a number: 51
It is not a Palindrome.
```